Forecasting AMD Stock Prices Using ARIMA: Evidence of a Random Walk with Drift

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Regression Analysis for Social Sciences 2

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Introduction

The stock market plays a crucial role in global economic activity. Accurate predictions of stock prices are beneficial for academic research, national country policies, and investment decisions. This study focuses on historical price patterns of Advanced Micro Devices, Inc. (AMD) to forecast short-term price movements using time series models. Although understanding the irrational behavior of stock prices remains a challenge, time series models such as ARIMA can offer reasonable short-term forecasting performance.

Angadi and Kulkarni (2015) applied time series models such as ARIMA in R to forecast stock prices, showing that these methods can effectively capture short-term trends. Recent study by Sunki et al. (2024) compared ARIMA with LSTM models and found that while ARIMA performs well in the short term, LSTM offers better long-term accuracy. These studies support the use of ARIMA as a practical model for short-term stock price prediction.

The ARIMA(0,1,0) model with drift gave a clear and useful short-term forecast, showing an upward trend in AMD stock prices over the period.

Model and Describe Your Approach

This study applies the Autoregressive Integrated Moving Average (ARIMA) model of time series regression to forecast short-term stock price movements in AMD. Specifically, it estimates an ARIMA(0,1,0) model with drift, which assumes that the stock price follows a random walk with drift. This is a common approach for financial time series that exhibit strong persistence and potential non-stationarity. The model is expressed as:

$$yt = yt_{-1} + c + \varepsilon t$$

- yt is the closing stock price at time t.
- c is the drift term.

- εt is a white noise error term, assumes this has zero mean and constant variance

This equation implies that today's price equals yesterday's price plus a constant average change (drift) and a random shock.

The selection of the ARIMA(0,1,0) with drift model was based on standard time series analysis procedures. A stationarity check using the Augmented Dickey-Fuller (ADF) test showed that the closing stock price series is non-stationary in level. After first differencing, it becomes stationary, which sets the integration order to d=1. No significant autocorrelation was found in the residuals of the model, which supports setting the autoregressive and moving average terms to zero (p=0, q=0). This specification implies that the data follows a random walk with drift.

In the ARIMA(0,1,0) with drift model, the key parameter of interest is the drift term. It represents the expected average change in the value of the time series over time. After transforming the data into period-to-period differences, the drift term indicates whether there is a consistent upward or downward trend. For the expected signal, a positive drift is generally expected, especially for a growing company like AMD, as it reflects a long-term upward trend in price.

Firstly, to estimate the ARIMA(0,1,0) with drift model, the Augmented Dickey-Fuller (ADF) test is used to assess stationarity. Since the original series is non-stationary, the data is different once to achieve stationarity. After differencing the series, diagnostic checks are performed to confirm that no autoregressive or moving average terms are needed, and a drift term is appropriate. Parameters are estimated using maximum likelihood estimation (MLE). After estimation, residual tests are conducted to ensure the model is well-specified and suitable for forecasting.

Empirical Analysis

The data used in this study was collected from Yahoo Finance and consists of daily closing prices for AMD over a two-year period, from July 1, 2023, to June 30, 2025. Daily closing prices were used

based on standard practice in financial forecasting and are further supported by previous research, such as the study by Angadi and Kulkarni (2015), which highlights their effectiveness in capturing daily market trends. Furthermore, the observation range was selected to ensure a sufficiently large number of data points to improve the reliability and performance of time series forecasting models, as recommended by Box and Tiao (1975).

Firstly, the Augmented Dickey-Fuller (ADF) test was applied to check for stationarity in the original series. The test returned a p-value of 0.6517, indicating that the series is non-stationary at level. To address this, the series was differenced once, and the ADF test was applied again, resulting in a p-value less than 0.01, which confirms that the differenced series is stationary.

```
        Augmented Dickey-Fuller Test

data:  price
Dickey-Fuller = -1.8265, Lag order = 7, p-value = 0.6517
alternative hypothesis: stationary


        Augmented Dickey-Fuller Test

data:  dprice
Dickey-Fuller = -7.5286, Lag order = 7, p-value = 0.01
alternative hypothesis: stationary
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Table 1. Results of Augmented Dickey-Fuller Tests for AMD Stock Price Series. Top is level price, bottom is applied to the first differenced series.

An ARIMA(0,1,0) model with drift was then fitted to the data using maximum likelihood estimation (MLE). As shown on Table 2, the estimated drift term was 0.0562, which reflects a positive average daily change in the stock price. This suggests an overall upward trend over the period. The model selection is supported by residual diagnostics, such as the Ljung-Box test, which returned a p-value of 0.8955. This number indicates no significant autocorrelation in the residuals. Additional diagnostic checks ensure the residuals are approximately normally distributed and homoskedastic.

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ARIMA(0,1,0) with drift

Coefficients:
       drift
      0.0562
s.e.  0.1872

sigma^2 = 17.48:  log likelihood = -1418.57
AIC=2841.15   AICc=2841.17   BIC=2849.57

Training set error measures:
                      ME     RMSE      MAE         MPE     MAPE      MASE        ACF1
Training set 0.0002319915 4.172857 2.965312 -0.04797282 2.215929 0.9981043 -0.02207857
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Table 2. Estimation Results and Forecast Accuracy of ARIMA(0,1,0) with Drift

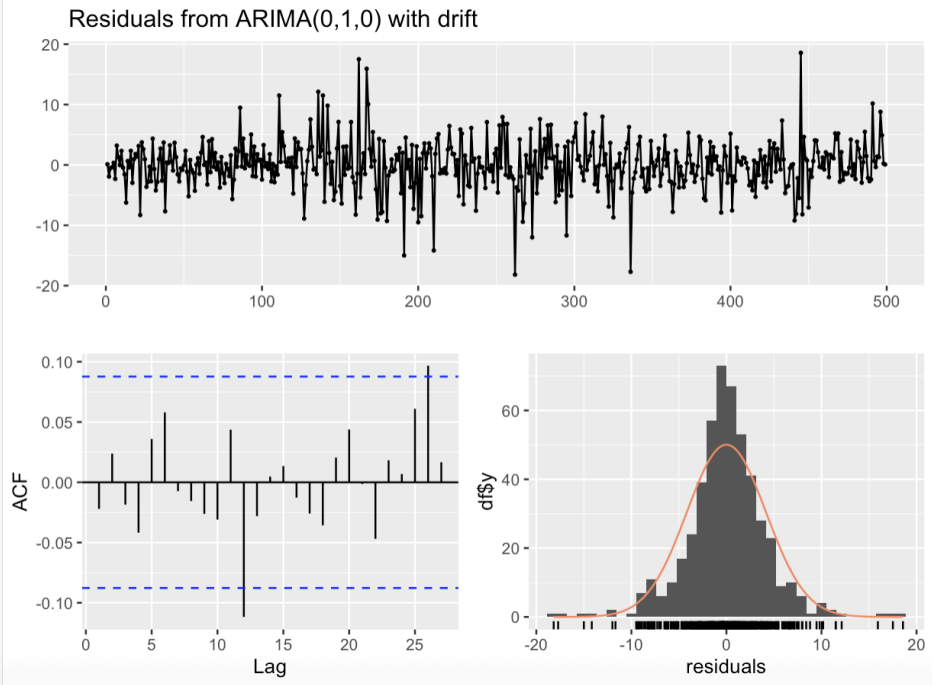
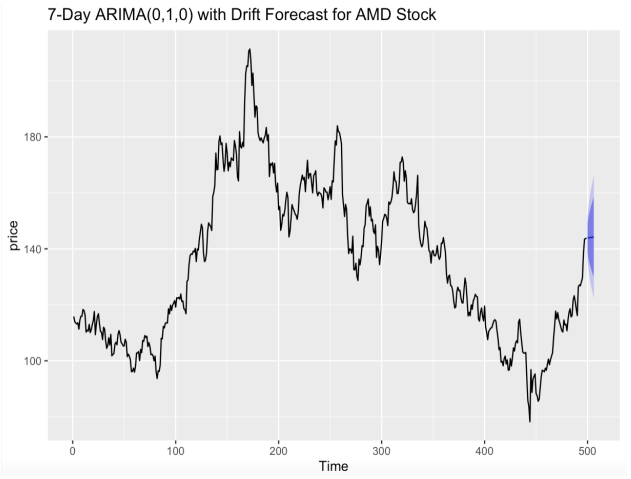


Table 3. Diagnostic Plots for Residuals from ARIMA(0,1,0) with Drift

The model is expected to forecast the future 7 days of AMD stock closing prices. The Table 4 shows slightly upward, with a positive drift term. In Table 5, the forecast for Day 1 is approximately 143.87, and by Day 7 it rises to 144.20. The blue shaded area shows the 95% confidence interval, which gets wider over time. This means the uncertainty increases further if try to predict the future. The result shows that the model works well for short-term forecasts, but stock prices become harder to predict as time goes on.



Forecasts:

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
500	143.8662	138.5077	149.2247	135.6711	152.0613
501	143.9224	136.3444	151.5004	132.3328	155.5120
502	143.9786	134.6975	153.2598	129.7843	158.1729
503	144.0348	133.3179	154.7518	127.6446	160.4250
504	144.0910	132.1091	156.0729	125.7662	162.4158
505	144.1472	131.0217	157.2728	124.0734	164.2210
506	144.2034	130.0262	158.3806	122.5213	165.8856

Table 4 &5: Graph and Summary of 7-Day Forecast for AMD Stock (ARIMA(0,1,0) with Drift)

This model comes with Root Mean Squared Error (RMSE) of 4.17 and Mean Absolute Error (MAE) of 2.97, reflecting reasonable short-term predictive accuracy from Table 2.

Conclusion

This study shows that the ARIMA(0,1,0) model with drift provides a reasonable and interpretable method for forecasting short-term stock price movements of AMD. Residual diagnostics confirm the model is well-specified, and the 7-day forecast shows reasonable accuracy. To conclude this study, ARIMA gives a reliable approach for short-term stock price prediction. In future studies, ARIMA(0,1,2) or ARIMA(0,1,1) could apply to do the estimate and comparison, to examine whether including additional lags in the moving average component improves the forecasting accuracy.

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